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BACHELOR'S THESIS

**Can we recover differential transcript
usage (DTU) from scRNA-seq?**

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Abstract

Single-cell RNA sequencing has transformed the study of cellular heterogeneity, yet widely adopted droplet-based protocols capture only the 3' end of each transcript [2]. This constraint makes Differential Transcript Usage (DTU) difficult to detect, as short terminal reads lack the internal splice junctions required to distinguish isoforms. Furthermore, proportion-based DTU methods—including those extended to single-cell data, which rely on pseudo-bulk aggregation—sacrifice cell-level resolution to recover the read depth they require [3, 4].

This thesis investigates whether COTAN—a statistical framework that models zero counts directly through presence/absence contingency tables [1]—can be extended from gene-level co-expression to transcript-level DTU detection. By bypassing both artificial imputation and pseudo-bulk aggregation, the adapted method identifies mutually exclusive isoform pairs via independence testing and bidirectional contrast thresholds across cell clusters.

A reproducible Nextflow pipeline was built to construct transcript-level matrices from raw 3'-biased data using alevin-fry with identity transcript-to-gene mapping [5]. This pipeline was applied to a human cell line mixture (29,000 cells) and mouse brain tissue (4,000 cells). Transcript-level clustering successfully recovered fine-grained substructures consistent with the Global Differentiation Index, and top DTU candidates—including Meg3, Malat1, and TPM1—were consistent with documented isoform biology [6].

These results demonstrate that sparsity-robust, contingency-table statistics can detect isoform mutual exclusivity in droplet-based data without the prohibitive cost of full-length sequencing. While sensitivity remains bounded by positional bias—rendering mid-transcript splicing events invisible—and orthogonal validation is still required, this work establishes a principled path for isoform-aware analysis on large-scale atlas datasets.

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Glossary

This glossary defines the core concepts whose discussion spans several chapters. Terms needed only once are defined inline as footnotes where they first appear; the entries below are those the reader may want to revisit. Each term is defined here and hyperlinked from its first mention in the text.

3' end The terminal end of an RNA molecule at which transcription finishes; in mRNA, it is where the poly-A tail is added. In 3'-biased protocols, the capture chemistry is anchored at this end, so reads cover only the terminal fragment of each molecule rather than the whole transcript. 3, 39

3'-biased sequencing A property of some scRNA-seq protocols (such as 10x Genomics Chromium) in which sequencing reads are derived only from the 3' terminus of each transcript. Because reading quality decreases by distance and because capture is anchored at the transcript's 3' end, coverage does not extend into the transcript body, so internal splice junctions are never observed [7]. 4, 13, 15, 23

alternative splicing The process by which a gene's long primary RNA copy is shortened and rearranged during maturation. The copy contains coding blocks (exons) and non-coding blocks (introns); splicing removes the introns and joins the flanking exons at the resulting splice junctions. When exon selection varies between transcripts of the same gene, the result is multiple distinct isoforms. 1, 20, 24

bulk RNA-seq Sequencing of RNA pooled from many cells at once, which averages away cell-to-cell differences. 2, 4, 17

cell barcode A short DNA tag carried by every read from a given cell, identifying its cell of origin. It allows the reads of one pooled sequencing run to be regrouped into per-cell expression profiles. 2, 10, 15, 23

cell cluster A group of cells with similar expression profiles, obtained by clustering the count matrix. Most single-cell analyses compare cell clusters against one another. 5

co-expression The tendency of two features (genes or transcripts) to be detected together across cells. 27, 30

contingency table A 2×2 table counting, across all cells, the four presence/absence combinations of two features; it tests whether they co-occur more or less than chance. 17, 27, 30

count matrix A table whose rows index features (genes or transcripts) and whose columns index cells, with each entry recording how many molecules of that feature were detected in that cell. 3, 4, 8, 24

differential gene expression (DGE) The analysis of differences in a gene's total expression—the summed expression of all its isoforms—between groups of samples. It measures overall gene output and is blind to changes in the relative usage of individual isoforms, which are instead captured by differential transcript usage. 3, 20

Differential Transcript Usage (DTU) Differential transcript usage occurs when the relative expression of multiple transcripts arising from the same gene changes between different conditions [8]. It is independent of changes in the gene's total expression, so it is not captured by gene-level differential expression analysis and requires transcript-level resolution to detect. 1, 7, 20, 21, 29

droplet-based protocols (droplet data) High-throughput single-cell protocols that encapsulate each cell together with a barcoded bead inside a picolitre droplet, so that tens of thousands of cells are processed in one run. They are the dominant scRNA-seq format, and their capture chemistry reads only the tail of each transcript. 3

FASTQ The plain-text de facto standard output of Illumina sequencers, named by analogy to FASTA with a Q for quality. Files are almost always gzip-compressed, taking the .fastq.gz extension. 10, 24

full-length sequencing A sequencing approach that reads an RNA molecule from one end to the other, rather than only the fragment nearest one of its ends. 2, 5, 8

gene A functional unit of the genome that specifies one or more RNA molecules. 1, 24

genome The complete set of genetic material of an organism, written in DNA and inherited from one generation to the next. It is organised into genes, the functional units whose expression is measured. 1

Global Differentiation Index (GDI) A cluster-specificity measure used to identify marker genes and validate clustering quality. 19

isoform A distinct mature transcript variant produced from the same gene by alternative splicing. 1, 3, 4, 24

isoform switching A change between conditions in the relative usage of a gene's isoforms, such that the fraction of the gene's expression contributed by one isoform increases while that of another decreases. It reflects altered alternative splicing or transcription and is quantified by the change in each isoform's fraction of total gene expression [9]. 4, 21, 37

library preparation The wet-lab step that converts captured RNA into sequencer-ready DNA: reverse transcription to cDNA, fragmentation or tagmentation, and adapter ligation that adds the sequencing primer sites and an index barcode. 9

messenger RNA (mRNA) The protein-coding subtype of RNA that carries a gene's sequence to the ribosome. 1

mutual exclusivity Two features that rarely appear together in the same cell (a negative association), as opposed to co-expression. 21, 30, 37, 41

orthogonal validation Independent confirmation of a computational result by a different experimental technique. 37

PCR Polymerase chain reaction: the amplification step that copies each captured cDNA fragment many times over, so that the few molecules recovered from a cell are present in enough quantity for the sequencer to read [10]. Because copies outnumber original molecules, the UMI tag is what allows them to be collapsed back to a single count. 2, 10, 25

poly-A tail A chain of adenine bases appended to the 3' end of a mature mRNA; it stabilises the molecule and serves as the binding site for capture primers in tail-primed protocols. 9

protein A molecule built from a chain of amino acids that performs the work of the cell, including catalysing reactions, transmitting signals, and forming structural components. 1

pseudo-bulk aggregation The practice of summing read counts across groups of single cells to simulate bulk RNA-seq depth, which recovers statistical power at the expense of single-cell heterogeneity. 4

read A short nucleotide sequence produced by the sequencing machine, representing the raw data of RNA-seq. 2

RNA (ribonucleic acid) The broad class of nucleic acid molecules transcribed from DNA. It comprises messenger RNA (mRNA), which carries protein-coding instructions, and non-coding RNAs that perform structural or regulatory roles. 1

RNA sequencing (RNA-seq) A technique that determines the sequence and abundance of RNA molecules in a sample; it is the basis of the single-cell protocols discussed in this thesis. 2, 41

single-cell RNA sequencing (scRNA-seq) RNA sequencing performed on thousands of individual cells separately, exposing per-cell expression. 2, 7, 8, 24

sparsity The very high proportion of zero counts in single-cell RNA-seq data. 3, 8, 9

splice junction The boundary created where two coding blocks (exons) are joined after the intervening non-coding block (intron) has been removed. Reads spanning a splice junction identify which blocks were joined together. 1

tag bias A bias in which each molecule is read only from one of its ends, so that coverage concentrates near that end while the interior of the transcript is never observed. It follows from a capture chemistry anchored at a single end, as in the tail-primed droplet protocols. 2, 3, 9, 10

technical dropout A zero count produced by a technical limitation of the assay—failed capture, reverse transcription, or sequencing—rather than by the true absence of the transcript. It is also described as a non-biological zero. 10

transcript An RNA molecule copied from a gene. 1

Unique Molecular Identifier (UMI) A short random sequence attached to each captured RNA molecule before PCR amplification, so that amplified copies of one molecule are recognised and counted only once. 2, 10, 15, 25

unspliced/spliced transcript The first RNA copy of a gene, still containing its non-coding blocks (introns); it is also called the primary transcript or pre-mRNA. Its counterpart is the spliced transcript, the same molecule once those blocks have been removed, which is the mature form translated into protein. 1, 27

Chapter 1

Introduction

Biological Context for Computer Scientists

Proteins are the basic building blocks of a cell and its DNA *genome* encodes the information needed to build them. Every cell of a multicellular organism carries essentially the same DNA, yet cells develop into distinct types. The difference cannot come from the DNA itself. Between the reading of the genome and the building of proteins, therefore, lies one or more layers of regulation that determine which proteins a cell produces, and those proteins in turn give the cell its characteristics. Characterising these layers is central to explaining how cells function and what shapes their behaviour.

Gene expression proceeds in three stages. A *gene* is first copied into a long molecule of *RNA (ribonucleic acid)*, called *unspliced transcript*. The second stage, *alternative splicing*, removes the non-coding blocks and joins the coding ones at the resulting *splice junctions*, yielding a shorter spliced *transcript*. The final, spliced version is *messenger RNA (mRNA)*, which the ribosome translates¹ into protein in the third and final phase.

The shortening process is not a one-to-one mapping. To expand the decoding capacity of a gene, cells can produce several different spliced versions of the same *unspliced transcript*; these versions are called *isoforms*. Because different isoforms can translate into proteins with entirely different, or even conflicting, roles, a cell can fundamentally alter its behaviour simply by switching which variant it expresses—the MET Δ ₁₄ isoform, for instance, acts as a switch that drives invasive growth in cancer [11]. The core goal of *Differential Transcript Usage (DTU)* analysis is therefore to measure changes in the relative expression proportions of specific transcript isoforms from the same gene across different biological conditions, developmental stages, or cell types, in order to expose regulatory events that gene-level measurement renders invisible.

¹ **Translation:** The process by which the ribosome reads an mRNA molecule and assembles the corresponding chain of amino acids into a protein.

A Brief History of Single-Cell Sequencing

For most of its history, *RNA sequencing (RNA-seq)* was performed on bulk samples. RNA is extracted from a population of cells, pooled into a single reaction, and sequenced as one library², so the resulting profile describes the sample as a whole rather than any cell within it. Sequencing of this kind, termed *bulk RNA-seq*, is statistically comfortable: pooling thousands to millions of cells yields deep coverage and stable abundance estimates for every gene. The averaging, however, is also its principal limitation. A gene's measured abundance is the mean over all cells present, so cellular heterogeneity³ disappears before the data ever reach an analyst. A rare cell type constituting two percent of a tissue contributes two percent of that tissue's *reads* for each of its markers⁴ and is easily buried beneath the dominant population; two subpopulations that shift in opposite directions—one gaining a transcript while the other loses it—cancel out and appear unchanged.

Single-cell RNA sequencing (scRNA-seq) replaced that aggregate view with per-cell profiling, running the same measurement independently on thousands of cells. The strategy rests on compartmentalisation. Each cell is physically isolated, its polyadenylated RNA captured and reverse-transcribed separately, and every resulting molecule marked with two labels. A *cell barcode* identifies the cell of origin, so reads can be regrouped into per-cell profiles after sequencing; a *Unique Molecular Identifier (UMI)* tags the individual molecule, so *PCR (polymerase chain reaction)* duplicates are collapsed rather than miscounted as expression. A single pooled sequencing run is thereby converted into thousands of independent expression measurements, each far shallower than a bulk sample.

The first *scRNA-seq*, produced from 2009 onwards, came from plate-based protocols, which consume one cell per reaction [12]. The chemistry of that era anchored its reads at one end of the molecule, and base accuracy declines along a read as it lengthens.⁵ Reliable information therefore concentrated near that end, a limitation that deeper sequencing could not remedy—data of this kind exhibit *tag bias* [13].

In 2012, the invention of *full-length sequencing* removed that bias [14, 15]. A whole molecule could now be copied and read from end to end, preserving the internal structure that distinguishes one transcript variant from another. While this was a substantial gain in coverage, the technology was costly and the plate format kept a study at hundreds to a few thousand cells.

² **Sequencing library:** The set of DNA molecules loaded into the sequencer for one run. Here it holds the sample's RNA, copied into DNA and tagged so that the instrument can read each fragment.

³ **Cellular heterogeneity:** The coexistence of distinct cell types and states within a single organism.

⁴ **Marker genes:** A gene or transcript whose expression specifically identifies a cell type, state, or condition.

⁵ **Base:** One of the four nucleotides that form a DNA or RNA strand: adenine (A), cytosine (C), guanine (G), and thymine (T). A read is a sequence of consecutive bases, and the confidence with which each base is identified declines along a read as it lengthens.

A new protocol, droplet microfluidics, suddenly revolutionised the scale in 2015 [16]: each droplet encapsulates one cell together with a bead⁶ carrying barcoded primers⁷, so tens of thousands of cells are profiled in one run at a small fraction of the cost per cell. The gain came at a price. Droplet capture is anchored at the transcript tail; this forces the use of the old chemistry, and the tag bias returned.

Droplet protocols dominate single-cell work today, since they profile far more cells per experiment at a far lower cost per cell than plate-based chemistry allows [2]. What tag bias removes is therefore the central limitation of the format rather than an incidental one, and recovering it is an active line of work. Long-read platforms read whole transcripts at single-cell resolution, and short-read datasets are mined for whatever isoform signal their terminal fragments still carry [6].

The Computational Challenge

Two obstacles stand between droplet data and transcript-level results.

Extreme sparsity. *Count matrices* from droplet experiments are mostly zeros, with zero entries routinely forming 70% to 90% of all values. Some of those zeros are biological, recording a transcript that the cell never produced, while others are technical, recording a molecule that was present but never captured, and both look identical in the matrix.

Ambiguous reads. Only the terminal 3' end of each molecule is read, so where several *isoforms* share that end they look identical in the data and their proportions cannot be measured directly.

Deeper sequencing reduces the first obstacle and cannot touch the second: extra reads fill in more of the matrix, but no quantity of terminal fragments recovers the interior of a transcript. A method that works on droplet data must therefore extract signal from a lightly observed count matrix while accepting that most of each molecule was never seen.

A Note on What Remains Recoverable

Gene-level information survives the *tag bias* constraint. A gene registers when a read is confidently attributed to it, and in most cases a read that cannot be traced to a single transcript still maps to isoforms of the same gene. A gene matrix can therefore be built from the same reads, and it remains informative for *differential gene expression (DGE)* analysis. A consequential property is that this matrix is denser than the transcript matrix.

⁶ **Bead:** A small gel sphere loaded with the barcoded primers. One bead is co-encapsulated with each cell, supplying every primer its droplet needs.

⁷ **Barcoded primers:** Primers that carry a short identifying sequence, so that every molecule captured in a given droplet can be traced back to its cell after sequencing.

Current Approaches and Limitations

Three families of methods address transcript-level regulation, and each stops short of detecting DTU across a full droplet dataset.

Bulk algorithms. Established *bulk RNA-seq* tools such as rMATS [17], SUPPA2 [18], DRIMSeq [19] and DEXSeq [20] model continuous isoform proportions from deep, junction-spanning coverage. Applied to sparse droplet matrices, those assumptions break down and the resulting estimates are unstable.

Aggregation to pseudo-bulk. One route around the coverage problem is *pseudo-bulk aggregation*, which sums the cells of a cluster or condition into a single profile to recover the depth that bulk tools expect [3, 21]. The cost is the cell-to-cell variation the assay was built to measure: a switch confined to a small subpopulation is diluted by the cells around it.

Targeted 3'-biased sequencing tools. Sierra [4] analyses droplet data directly, but only for a narrow class of events at transcript termini. It offers no genome-wide framework for multi-*isoform switching* from raw zero-heavy counts.

What is missing is a framework that works on sparse droplet matrices at single-cell resolution, without collapsing cells together and without full-length reads.

Thesis Objective

The objective of this thesis is to test whether sparsity-robust statistical modelling can clear that bottleneck and detect DTU at single-cell resolution.

The work builds on COTAN (Co-expression Tables Analysis) [1], a framework for gene-level *count matrices* that focuses its statistical model on the zeros of such a matrix rather than on its positive counts. It asks whether two genes are detected together less often than chance would predict, using presence/absence contingency tables, so no artificial imputation⁸ or variance-stabilising transformation⁹ is required. That zero-centred construction performs well on the sparse gene matrices typical of single-cell data, which raises the question this thesis pursues: whether the same statistics, applied to transcript matrices instead, whose counts are sparser still, will hold. Applied one level lower, the same zeros would indicate which *isoform* of a gene a cell uses, and not only which genes it expresses.

⁸ **Imputation:** Replacing zero counts with values predicted from neighbouring cells, at the risk of inventing spurious correlations.

⁹ **Variance-stabilising transformation:** A transformation applied to counts, such as a logarithm, so that their variance no longer depends on their mean. It alters the scale of the data, turning an undetected molecule into a small positive value rather than a zero.

Three contributions follow:

Algorithmic adaptation. An isoform-aware framework identifies mutually exclusive transcript pairs within a gene, then applies bidirectional contrast filtering¹⁰ across *cell clusters* to isolate genuine switches.

A reproducible quantification pipeline. An automated, containerised Nextflow¹¹ workflow uses modified *alevin-fry*¹² indexing to derive transcript-level count matrices directly from raw 3'-biased reads, without gene-level collapse.

Empirical validation. The adapted framework is benchmarked on a homogeneous human cell line mixture and on complex mouse cortical tissue, cross-referencing candidate switches against published isoform biology and comparing transcript-derived clusters against gene-level baselines.

The thesis does not claim to circumvent the physical limits of tag bias: isoforms that share their terminal region remain indistinguishable. It maps instead the practical boundary of computational recovery, asking where sparsity-robust contingency statistics recover differential transcript usage across large droplet datasets without the cost of *full-length sequencing*.

¹⁰ **Bidirectional contrast filtering:** Retaining a transcript pair only when at least one cluster favours one transcript and at least one other cluster favours the other, which separates a genuine switch from one-sided enrichment.

¹¹ **Nextflow:** A dataflow workflow language: a pipeline is declared as processes connected by typed channels, and an execution engine schedules the resulting directed acyclic graph of tasks across the available cores or machines. Each process runs in its own container, so the pipeline is portable between systems and reproducible from its container images rather than from the host environment.

¹² **alevin-fry:** A fast single-cell quantification tool that assigns reads to transcripts by pseudo-alignment.

Chapter 2

Theory and Frameworks

This chapter provides the theoretical foundation for understanding the methods in Chapter 3. It covers *scRNA-seq* preprocessing concepts, the COTAN framework for sparse data [1], and *DTU* fundamentals.

Notation

Mathematical notation is used in this chapter and the two that follow only where it clarifies an argument. The table below collects the symbols that recur across them; symbols confined to a single local derivation are introduced where they appear.

Symbol	Meaning
X	Expression count matrix produced by the preprocessing pipeline, with features (transcripts or genes, depending on the pipeline output level) in rows and cells in columns
x_{fc}	Observed integer count of feature f in cell c
f, c	Feature and cell indices
N	Number of cells
R	Total number of sequencing reads
L	Sequenced read length
M	Mean number of reads per cell
k	Length of the k -mers used to index the reference
$ \Sigma $	Alphabet size (4 for DNA)
T	Number of annotated transcripts in the reference
T_{obs}	Distinct transcripts detected in a single cell
L_{avg}	Mean transcript length
θ_t	Relative abundance of transcript t
ℓ_t	Effective length of transcript t
C_r	Equivalence class of transcripts compatible with read r
$\hat{z}_{rt}$	Fractional assignment of read r to transcript t

Symbol	Meaning
<i>Statistics and detection</i>	
a_c, b_c	Binary presence indicators for a pair of features in cell c
O_{ij}, E_{ij}	Observed and expected counts in a 2×2 contingency table
Row $_i$, Col $_j$	Marginal totals of that table
χ^2	Pearson chi-squared statistic
$C_{f,f'}$	Co-expression coefficient between features f and f'
$p_{f,f'}$	Independence p -value for that pair
$D_{f,S}$	Differential expression (DEA) coefficient of feature f in cluster S
S	Cell cluster
τ	Contrast threshold on the DEA difference within a gene
gdi_i	Cluster-specificity index of feature i

Preprocessing Pipeline

This section follows a single *scRNA-seq* experiment from its data-generating mechanism to the *count matrix* consumed by every later analysis. Its scope is the route this thesis actually takes—droplet chemistry, pseudo-alignment, expectation-maximization—with the alignment alternative noted only where it changes the outcome. Plate-based and alignment pipelines differ in the details of each phase, not in the phase structure. It treats that path as a four-phase pipeline: ingestion and quality control, mapping, probabilistic resolution of ambiguous reads, and deduplication into a matrix. Each phase is mathematically and biologically constrained by the one before it, so the physical hardware chosen in the initial step dictates the shape, *sparsity* and content of the final analytical matrix.

Phase 0: Choosing the Protocol

The design of a *scRNA-seq* analysis pipeline is fundamentally constrained by the physical protocol chosen during data generation. Rather than representing a computational phase, this hardware choice establishes the data’s absolute boundaries, dictating the scale, resolution, and inherent *sparsity* of the resulting *count matrix*. The available protocols bifurcate into two categories with distinct computational trade-offs: plate-based methods and droplet microfluidics.

Plate-based methods, such as SMART-seq2¹, physically isolate individual cells into the discrete wells of a 96- or 384-well plate. This architecture supports a capture chemistry capable of *full-length sequencing*: each transcript is captured and converted to full-length cDNA² across its entire length, preserving the internal splice junctions necessary for direct isoform differentiation. Because each well constitutes an independent

¹ **SMART-seq2:** A plate-based full-length *scRNA-seq* protocol, one cell per well, capturing whole transcripts.

² **cDNA (complementary DNA)** is a DNA copy of an RNA molecule. Sequencers read DNA, not RNA, so the RNA must first be reverse-transcribed into cDNA before it can be read.

library preparation, throughput is strictly bounded by reagent costs and processing labour, typically constraining experiments to a few thousand cells. In exchange, each cell receives substantial sequencing depth, yielding highly detailed per-cell profiles that make the data highly informative for isoform-level questions [2].

Conversely, droplet microfluidics replace physical wells with self-assembling picolitre³ droplets. Platforms such as the 10x Genomics⁴ Chromium system co-encapsulate single cells with barcoded beads, scaling throughput to tens or hundreds of thousands of cells per run at a fraction of the cost per cell [22]. This massive parallelisation imposes severe structural constraints on the data. Capture relies on oligo-dT⁵ primers anchored to the *poly-A tail*, restricting sequencing strictly to the 3' terminus⁶ of each transcript. This *tag bias* intrinsically discards the internal sequence structure required to resolve complex splicing events. Furthermore, a finite sequencing capacity distributed across a massive cell population drastically reduces the reads allocated per cell to a few thousand molecules, while the stochastic nature of droplet encapsulation necessitates rigorous downstream algorithmic filtering to remove empty droplets and multi-cell multiplets⁷

Table 2.2 sets the two chemistries side by side.

Table 2.2: The two dominant single-cell library chemistries. Per-cell depth and cost trade off against throughput; total reads per day are set by the sequencer, not the chemistry. All figures describe a single sequencing run. Values are order-of-magnitude, after Ding et al. [22].

	Plate (SMART-seq2)	Droplet (10x)
Library preparation	~ 2 days for 96 cells	~ 1 day for 10^4 – 10^5 cells
Cells per experiment	10^2 – 10^3	10^4 – 10^5
Reads per cell	~ 10^6	~ 10^4
Detected transcripts per cell	~ 10^4 – 10^5	~ 10^3
Reads per day	~ 10^8 – 10^9	~ 10^8 – 10^9
Coverage	full transcript	3' terminus only
Reagent cost per cell	~ 10–100× higher	baseline

The defining computational consequence of the droplet architecture is extreme matrix *sparsity*, which follows directly from comparing the size of the reference with what a single cell yields. Let T denote the number of annotated isoforms in a standard mammalian reference and T_{obs} the distinct transcripts successfully recovered per single cell. While

³ **Picolitre:** One trillionth of a litre (10^{-12} L), roughly the volume of a sphere about 12 μ m across—comparable to a single cell.

⁴ **10x Genomics Chromium:** The dominant commercial droplet platform, barcoding each cell in a microfluidic droplet.

⁵ **Oligo-dT priming:** Capture primers made of a short chain of thymine bases, which anneal to the poly-A tail and start synthesis there. The captured fragment is therefore the 3' end of the molecule.

⁶ **3' (three-prime) terminus** is the end of a transcript at which the poly-A tail is added; capture chemistry anchored here reads only the terminal fragment, not the transcript body.

⁷ **Multiplet:** A droplet that receives more than one cell, yielding a mixed profile that quality control must remove.

the reference dictates $T \sim 10^5$, a typical droplet-sequenced cell yields only $T_{obs} \sim 10^3$. In the raw alignment space, this disparity produces an initial matrix zero rate of roughly 99%. Standard preprocessing pipelines subsequently filter out features that are entirely unexpressed or ultra-rare in the sampled tissue, condensing the effective feature space down to $T_{filtered} \approx 1.2 \times 10^4 - 2.3 \times 10^4$ viable transcripts. Within this practical, filtered domain, the observed zero rate consistently settles between 70% and 90%.

Some of these zeros are biological zeros⁸, recording a transcript the cell never produced; the rest are *technical dropouts*, recording a molecule that was present but never captured. Because the cumulative chemical efficiency of droplet capture, reverse transcription and sequencing succeeds on only 10% to 20% of the molecules present [23], the second kind dominates, and the two conflate into the 70% to 90% zero rate reported above. They are numerically identical once recorded, so no count-based test can separate them. The 3' *tag bias* deepens this sparsity further by collapsing distinct isoforms that share terminal exons into a single observational unit, and it is that overwhelmingly zero-heavy structure which dictates the need for a specialised statistical model in every downstream analysis.

The economics explain the adoption: a plate experiment consumes one well per cell at one to five dollars, while droplet encapsulation profiles a cell for a few cents, so a fixed budget buys two to three orders of magnitude more cells [2]. That trade is made knowingly. For the dominant questions—cell-type composition, marker discovery, differential gene expression—the payoff is statistical, because a per-gene test averages over thousands of cells and tolerates a sparsity that remains severe within any single one. Coverage loss is accepted wherever the question is gene-level, and resurfaces as a limitation only where it is isoform usage.

Phase 1: Data Ingestion and Quality Control

The first computational challenge in a single-cell pipeline is purely one of scale and structure. Sequencers output raw data in *FASTQ* format, representing each read across four distinct lines: a coordinate identifier⁹, the nucleotide sequence¹⁰, a separator¹¹, and the corresponding Phred-scale quality scores¹². For droplet-based chemistries such as the 10x Genomics platform, this raw sequence carries critical multiplexing metadata that must be carefully isolated before any biological analysis begins. Specifically, the sequencer must extract a *cell barcode* that maps the read back to its discrete cell of origin, and a *UMI*—a random tag attached prior to *PCR* amplification used to track individual captured mRNA molecules.

⁸ **Biological zero:** A zero count reflecting genuine absence of a transcript in a cell, as opposed to a technical dropout.

⁹ **Read identifier:** A line beginning with @ that names the read and records the instrument, flow cell and coordinates it came from.

¹⁰ **Nucleotide sequence:** The read itself, a string of A, C, G and T bases in the order the sequencer read them.

¹¹ **Separator:** A line beginning with + that marks the end of the sequence and may repeat the identifier.

¹² **Per-base quality score:** One character per sequenced base, encoding the probability that the base was called incorrectly on the Phred scale.

Before the reads can be quantified, they must be cleaned. Tools such as cutadapt¹³ or fastp¹⁴ trim synthetic adapters, discard low-quality bases, and extract the barcode and UMI payloads, while heuristic filtering separates valid cell barcodes from empty droplets and multi-cell collisions (multiplets). The step is optional, and is skipped under two conditions. When a vendor pipeline such as Cell Ranger has already trimmed adapters and corrected barcodes and UMIs, repeating the pass costs a full read of the FASTQ file without altering a base. When the mapper is read-structure-aware and UMI-aware, as alevin-fry and Salmon are, residual adapters fall outside the indexed k -mers and are never matched, so correction is left to the quantifier. Skipping is safe only under those guarantees: on untrimmed reads handled by a mapper that does not correct UMIs, adapters mis-assign reads and sequencing errors inflate the number of distinct UMIs. For plate-based chemistry no barcode or UMI exists to extract and adapter content is low, so trimming reduces to a quality filter.

Because quality control touches every single nucleotide exactly once, it acts as a highly parallelisable streaming operation strictly bound by the total read count, R , and the read length, L . The computational time complexity is entirely linear, scaling as:

$$\mathcal{O}(R \cdot L)$$

The physical footprint, by contrast, is large. One uncompressed FASTQ record—a 100 bp read together with its identifier and quality line—occupies roughly 250 bytes, so a droplet experiment of $R \sim 10^9$ reads, the order of magnitude considered here, holds $\sim 2.5 \times 10^{11}$ bytes, about 250 GB before compression. Modern multi-core trimmers sustain throughputs on the order of 10^8 reads per minute, so this I/O-bound filtering resolves in a few hours on a single workstation.

Phase 2: Mapping Algorithms (Alignment vs. Pseudo-Alignment)

Quantifying transcript expression is fundamentally a large-scale assignment problem: mapping short sequencing reads back to their reference transcripts of origin. The computational workload is strictly governed by the total read count, which scales as the product of the cell population and the sequencing depth. This relationship is defined mathematically as:

$$R = N \cdot M$$

where R is the total read count, N is the number of cells, and M is the average number of reads per cell. Both bounds below are derived with R symbolic and instantiated only once, here. The two datasets analysed in this thesis total roughly 10^8 – 10^9 reads, so $R \sim 10^9$ is the scale used throughout; pooled multi-sample atlases reach 10^{10} – 10^{11} reads and scale every cost in this section by the same factor. The search space and query geometry impose additional dimensional constraints, defined by the mammalian reference transcriptome and the physical sequencer output respectively:

$$T \sim 10^5, \quad L_{avg} \approx 1500 \text{ bp}, \quad L \approx 90\text{--}150 \text{ bp}$$

¹³ **cutadapt**: A read-trimming tool that removes adapter sequences and low-quality bases.

¹⁴ **fastp**: A fast all-in-one read preprocessor that performs adapter trimming, quality filtering and reporting in a single pass.

where T denotes the total number of annotated isoforms (including unspliced pre-mRNA), L_{avg} the average transcript length in base pairs, and L the sequenced read length in base pairs. Read length is strictly a hardware constraint; in a 3' assay, Read 1 isolates the cell barcode and UMI, leaving solely Read 2 to capture the actual transcript sequence.

Traditional alignment algorithms¹⁵ (e.g., BWA [24], Bowtie2 [25]) evaluate this mapping base-by-base to derive exact genomic coordinates using a Burrows-Wheeler Transform¹⁶ index. Building this index requires a one-time computational investment proportional to the entire transcriptome size, scaling as $\mathcal{O}(T \cdot L_{avg})$. However, during the query phase, each of the R reads is localised through a search algorithm whose time cost grows logarithmically with the size of the reference T . Factoring in the algorithm's tolerance for sequencing errors, insertions, and deletions, this produces a total execution time of:

$$\mathcal{O}(T \cdot L_{avg}) + \mathcal{O}(R \cdot L \log T)$$

Searching the index does not scan the reference: backward search extends the matched suffix one base at a time, at constant cost per base. The logarithmic term above instead reflects the branch-and-bound over candidate loci that error tolerance requires, since a reference holding more transcripts offers more positions to test. That search penalty is what makes exhaustive alignment computationally prohibitive across the billions of reads generated in a single-cell experiment.

Modern droplet-based transcriptomics circumvents this bottleneck by utilising pseudo-alignment algorithms, such as Salmon, alevin-fry, or kallisto [26, 27, 5]. This approach rests on the critical mathematical observation that downstream quantification requires only the identity of the transcript of origin, not the exact base-level coordinates of the read. Consequently, pseudo-alignment replaces logarithmic sequence searches entirely with constant-time hash table lookups. Each read is fragmented into its overlapping k -mers¹⁷, and every k -mer is queried in a pre-computed table recording which transcripts contain it. Intersecting these sets rapidly yields an equivalence class¹⁸ of candidate transcripts compatible with the read [5].

The procedure executes in two distinct computational phases. First, during index construction, the reference transcriptome is decomposed into an index—typically a colored de Bruijn graph¹⁹ mapping each k -mer to the specific transcripts containing it. This incurs a time cost identical to traditional alignment, but the index is stored sparsely: only the k -mers actually present in the reference are represented, giving a spatial footprint of $\mathcal{O}(d)$ for d distinct observed k -mers. The dense worst case, $\mathcal{O}(k \cdot |\Sigma|^k)$ over the alphabet of size $|\Sigma| = 4$, is $4^{31} \approx 4.6 \times 10^{18}$ entries and is never materialised. Like traditional alignment, this index is built exactly once before any read is processed.

¹⁵ **Read alignment:** Base-by-base mapping of a read to the reference, giving exact coordinates at high computational cost.

¹⁶ **Burrows-Wheeler Transform (BWT):** A reversible text transform that allows fast substring search in a compressed reference index.

¹⁷ **k -mer:** A contiguous substring of length k ; reads are matched to transcripts by the k -mers they share.

¹⁸ **Equivalence class:** The set of transcripts that could have produced a read, since all contain the read's k -mers.

¹⁹ **Colored de Bruijn graph:** A graph of overlapping k -mers in which each k -mer is coloured by the transcripts containing it, used as the pseudo-alignment index.

Second, during the query phase, each read is fragmented into its $L - k + 1$ overlapping k -mers. The algorithm executes a hash table lookup for every single k -mer, retrieving and intersecting their corresponding transcript sets to produce the equivalence class. With a standard $k = 31$ and $L \approx 90\text{--}150$ bp, this requires roughly 60 to 120 lookups per read. By shifting the core operation from a base-by-base sequence search to a hash retrieval, the per-read query cost collapses to $\mathcal{O}(L)$, completely decoupling it from the massive size of the reference transcriptome. The total execution time for pseudo-alignment therefore simplifies to:

$$\mathcal{O}(T \cdot L_{avg}) + \mathcal{O}(R \cdot L)$$

Instantiating the two query terms at $R \sim 10^9$ reads of length $L \sim 10^2$ bp against $T \sim 10^5$ transcripts, where $\log_2 T \approx 16.6$:

$$R \cdot L \log_2 T \approx 10^9 \cdot 10^2 \cdot 16.6 \approx 1.7 \times 10^{12}, \quad R \cdot L \approx 10^9 \cdot 10^2 = 1.0 \times 10^{11}$$

Two separate effects account for the gap. Pseudo-alignment removes the asymptotic $\log_2 T \approx 16.6$ factor, which is worth roughly one order of magnitude on its own. The remainder of the reported 50- to 100-fold speed-up comes from the constant factor: retrieving a hash-table entry costs far less than an inexact string match. The two effects multiply, and together they turn a query phase measured in days into one measured in hours.

Phase 3: Probabilistic Resolution via Expectation-Maximization

Pseudo-alignment introduces a distinct computational challenge in the form of multi-mapping reads²⁰—reads whose k -mers match more than one transcript because those transcripts naturally share exonic sequences. A read falling entirely within a shared exon carries no intrinsic sequence information about which specific isoform it originated from, prompting the pseudo-aligner to return a set of valid candidates rather than a single definitive transcript. This ambiguity is profoundly exacerbated in *3'-biased* protocols, where short reads capture exclusively the terminal region and frequently match every single isoform sharing that specific terminal exon [1].

Discarding these ambiguous reads is statistically unviable. Because they form the overwhelming majority of alignments in *3'-biased* data, summarily dropping them would severely deplete the count matrix and artificially bias isoform proportions toward rare variants that happen to carry unique sequences.

Modern pseudo-aligners instead resolve this ambiguity by casting quantification as maximum-likelihood estimation over a generative model of the sequencing experiment [26]. Each fragment is assumed to be produced in two stages: a transcript t is first drawn with probability proportional to its relative abundance θ_t , and a start position is then drawn uniformly from the ℓ_t positions at which a fragment of the observed length can

²⁰ **Multi-mapping read:** A read compatible with more than one transcript because those transcripts share its sequence.

begin. The admissible count ℓ_t is the transcript's ²¹. Crucially, the transcript of origin is latent: pseudo-alignment reports only the equivalence class C_r of transcripts compatible with read r [27]. Marginalising over the unobserved origin gives the probability of observing that class under a given abundance vector:

$$P(C_r | \theta) = \frac{\sum_{t \in C_r} \theta_t \ell_t}{\sum_{t'} \theta_{t'} \ell_{t'}},$$

and the quantity to be maximised is the product of these terms over all reads:

$$\hat{\theta} = \arg \max_{\theta} \prod_r P(C_r | \theta), \quad \sum_t \theta_t = 1.$$

Because the origin of each read is unobserved, this likelihood is optimised by the Expectation-Maximization (EM)²² algorithm, which alternates between two phases until convergence.

In the E-step²³, the current estimate is used to compute the posterior probability that read r originated from transcript t , given that it originated within the class C_r :

$$\hat{z}_{rt} = \frac{\theta_t \ell_t}{\sum_{t' \in C_r} \theta_{t'} \ell_{t'}} \quad \text{for } t \in C_r,$$

and $\hat{z}_{rt} = 0$ otherwise. The effective-length term ℓ_t is not optional: it encodes the fact that a long transcript contributes proportionally more fragments at equal abundance, so omitting it biases $\hat{\theta}$ toward short isoforms. The algorithm then proceeds to the M-step, where each abundance is re-estimated as the total posterior weight assigned to its transcript:

$$\theta_t \propto \sum_{r: t \in C_r} \hat{z}_{rt},$$

normalised so that the values of θ_t sum to exactly one across the transcriptome.

Two properties of this iteration matter for the interpretation of the resulting counts. First, each round is guaranteed not to decrease the log-likelihood, so the estimates converge monotonically to a *local* optimum rather than a guaranteed global one [28]. The likelihood surface is frequently flat or multimodal, so implementations initialise θ uniformly, stop when the relative change in log-likelihood falls below a tolerance, and may return different solutions for different starting points. Second, the model is closed-world: every read is assumed to originate from an annotated transcript, so an incomplete reference, unannotated isoforms or uncorrected fragment biases propagate directly into $\hat{\theta}$. The estimate is consequently a point estimate without uncertainty, and downstream

²¹ **Effective length:** The number of fragment start positions a transcript can generate; approximately its length minus the fragment length, plus one. It corrects for the fact that longer transcripts yield more fragments at equal abundance.

²² **Expectation-Maximization (EM):** An iterative algorithm that estimates transcript abundances and resolves ambiguous read assignments jointly until they converge.

²³ **E-step / M-step:** The two alternating EM phases: distribute reads by the current abundance estimate (E), then re-estimate abundances (M).

methods that require variance—particularly the DTU tools discussed later—recover it by bootstrapping the quantification step [29].

Because the summation runs over equivalence classes rather than individual reads, many reads that share a compatible transcript set are handled by a single update. The computational cost therefore scales with the number of *distinct* classes, not with the number of reads, which is what makes transcriptome-wide EM tractable. Alevin-fry’s parsimonious variant exploits this further by collapsing transcripts that share an identical set of k -mers into a single group and running the EM updates over those groups, effectively degrading the problem to gene-level quantification where isoform resolution is impossible [5]. This is precisely the regime that $3'$ -biased data tends to occupy, and it is worth stating why it cannot be repaired by more data.

Remark (identifiability under $3'$ capture). Let t and t' be two isoforms of the same gene whose reads fall only in equivalence classes containing both, as happens when the shared terminal region covers their effective lengths. The E-step above then assigns posterior weight through the ratio $\theta_t \ell_t / (\theta_t \ell_t + \theta_{t'} \ell_{t'})$ alone. With $\ell_t \approx \ell_{t'}$ that ratio reduces to $\theta_t / (\theta_t + \theta_{t'})$, so the likelihood depends on the two abundances only through their sum and is flat along $\theta_t + \theta_{t'} = \text{const}$: every split that preserves the total explains the observed reads equally well. Identifiability is therefore bounded by the divergence of the terminal region, not by the estimator, the reference or the number of reads, since sequencing the same fragment more deeply cannot separate two transcripts that fragment cannot distinguish. This is the formal reason the transcript-level analysis of this thesis can recover only those switches whose isoforms differ at or near their $3'$ ends.

Phase 4: Deduplication and Matrix Generation

Once each read has been assigned to a transcript, the remaining task is to count molecules rather than reads. The protocol attaches a *UMI*—a short random sequence—to each captured molecule before amplification, so every read carries both a *cell barcode* identifying its cell and a UMI identifying its molecule. Because PCR copies inherit both tags, the reads descending from one molecule are indistinguishable duplicates, and collapsing each such group recovers the molecule counts that sequencing had inflated.

The collapse must be performed on barcode, UMI *and* assigned transcript together. A UMI holds only a few random bases, so within one cell two unrelated molecules can share a UMI by chance; grouping on barcode and UMI alone would merge them, whereas including the transcript keeps them apart whenever they originate from different features.

The result is the expression count matrix X (Table 2.3), whose entry x_{fc} records the number of distinct molecules observed for feature f in cell c . Amplification no longer distorts the counts, so the differences remaining between cells reflect biology and capture efficiency rather than how often each molecule happened to be copied.

Table 2.3: The four phases of the preprocessing pipeline and the object each produces.

Phase	Operation	Output
0	Protocol choice	Library chemistry (plate or droplet)
1	Ingestion and quality control	Cleaned reads carrying a cell barcode and a UMI
2	Mapping (alignment or pseudo-alignment)	Reads assigned to candidate transcripts
3	Probabilistic resolution (EM, pseudo-alignment only)	Transcript-level abundance estimates
4	Deduplication and matrix assembly	Count matrix X

Figure 2.1 summarises the four phases and records what each one costs. The losses compound rather than compensate: the chemistry in Phase 0 decides which part of each molecule is readable at all, so every later phase works from that partial view. Phase 2 discards base coordinates, and Phase 3 leaves transcripts that share their terminal region statistically inseparable, which is the constraint quantified at the end of that phase. The figure is the reason Chapter 3 has to build a transcript-level matrix from raw reads: gene-level matrices lose the isoform distinction that Phase 3 could not recover in the first place.

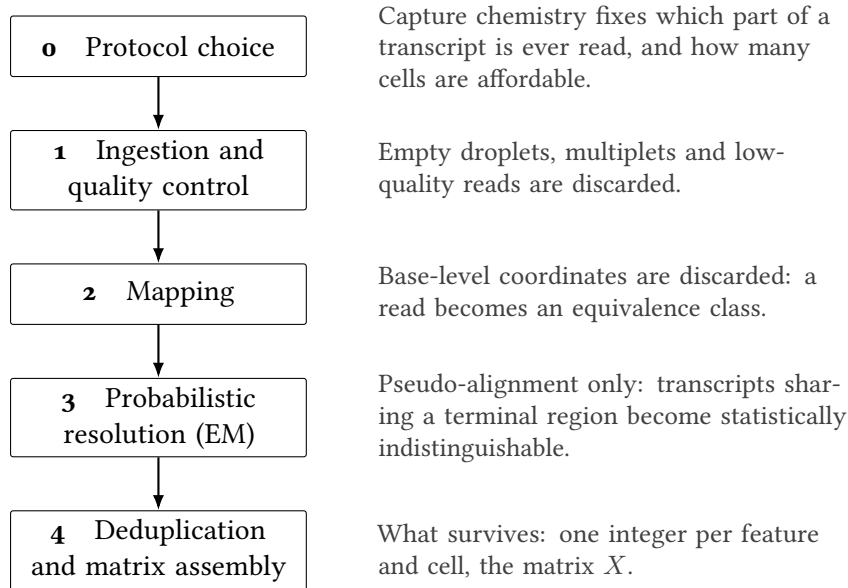


Figure 2.1: The four phases of the preprocessing pipeline and the information each one removes. Phase 3 exists only on the pseudo-alignment route, where ambiguous reads must be distributed probabilistically; direct alignment resolves every read by its coordinates and skips it. The chemistry chosen in Phase 0 bounds every later stage, so the losses accumulate rather than cancel.

COTAN Framework

The transcript-level matrix X produced above acts as the natural input for the statistical framework presented here, yet it simultaneously introduces the primary analytical difficulty: the shallow per-cell depth heavily imposed by the droplet hardware leaves the vast majority of its entries at zero. The COTAN framework, introduced by Galfrè et al. [1], was specifically engineered to analyse this zero-heavy structure directly, and has since been logically extended to recover entire cell clusters purely from co-expression patterns [30]. The following subsections formalise this sparsity and detail the contingency-table machinery COTAN employs to successfully bypass traditional density-based statistics.

The Sparsity Problem

Single-cell count matrices exhibit extreme structural sparsity, with 70% to 90% of all entries returning zero [1]. As noted when the matrix was introduced, the biological and technical origins of those zeros are indistinguishable once recorded: a transcript that was truly absent and one that was merely missed leave the same mark. That ambiguity, rather than the sparsity itself, is the fundamental statistical barrier for conventional tools designed for deep, dense *bulk RNA-seq* data.

Applying standard density-based statistical methods to such sparse matrices introduces severe, compounding computational artifacts. Log-normalisation, which strictly involves adding a pseudo-count before logarithmic scaling, heavily distorts low-abundance measurements and artificially compresses legitimate zeros into spurious scaling patterns [1]. Standard linear or rank-based association measures, such as Pearson or Spearman correlations, fail entirely in this domain, as they inevitably end up measuring the coincidence of technical dropouts rather than identifying true regulatory co-expression [1]. Furthermore, attempting to artificially densify the matrix through heuristic imputation—specifically replacing zeros with smoothed estimates derived from neighbouring cells—frequently introduces fabricated correlations and severely compromises sensitive downstream signals [1]. To fundamentally circumvent these pitfalls, COTAN explicitly abandons imputation and variance-stabilising transformations entirely, choosing instead to evaluate the zeros as a direct signal through strict binary discretisation.

Contingency Table Solution

COTAN cleanly bypasses the flawed continuous count amplitudes by evaluating pairwise feature dependencies using asymptotic independence testing applied strictly to binary presence and absence states [1]. For any given feature pair (f, f') across N cells, the continuous counts are logically reduced to discrete indicator variables for each cell c :

$$a_c = \mathbf{1}[x_{fc} > 0], \quad b_c = \mathbf{1}[x_{f'c} > 0]$$

Letting the discrete states be indexed by $i, j \in \{0, 1\}$, the framework constructs a comprehensive 2×2 *contingency table* where O_{ij} directly denotes the observed number

of cells exhibiting state $a_c = i$ and $b_c = j$. The marginal totals for this specific table are defined as $\text{Row}_i = \sum_j O_{ij}$ and $\text{Col}_j = \sum_i O_{ij}$, mathematically constrained by the total cell count $\sum_i \text{Row}_i = \sum_j \text{Col}_j = N$. Under the standard null hypothesis of statistical independence, the expected cell count E_{ij} is calculated directly as:

$$E_{ij} = \frac{\text{Row}_i \cdot \text{Col}_j}{N}$$

Table 2.4 lays the four counts out explicitly. The two diagonal cells hold cells in which the features are detected together (O_{11}) or neither is detected (O_{00}); the off-diagonal cells hold cells defining the difference between them, in which exactly one of the two is present. Co-expression resides in the tendency of the count to fall on a diagonal rather than on the anti-diagonal, and the expected value E_{ij} above is what that count would be under independence.

Table 2.4: The 2×2 contingency table built for a feature pair (f, f') across N cells. Entry O_{ij} counts cells with $a_c = i$ and $b_c = j$; the margins are the feature's detection totals.

	f' detected ($b_c = 1$)	f' absent ($b_c = 0$)	Row total
f detected ($a_c = 1$)	O_{11}	O_{10}	Row ₁
f absent ($a_c = 0$)	O_{01}	O_{00}	Row ₀
Column total	Col ₁	Col ₀	N

Deviation from this independent expectation is evaluated using Pearson's chi-squared statistic [1]:

$$\chi^2 = \sum_{i \in \{0,1\}} \sum_{j \in \{0,1\}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

The result is reported not as a statistic but as a single coefficient. COTAN defines the co-expression coefficient $C_{f,f'}$ by rescaling the deviation so that it lies in $(-1, 1)$ and, under the null hypothesis, satisfies $N C_{f,f'}^2 \sim \chi^2(1)$ over the N cells of the table. The pairwise p -value, the quantity used for screening throughout this thesis, then follows directly:

$$p_{f,f'} = P(\chi^2(1) > N C_{f,f'}^2)$$

A positive coefficient means the two features are detected together more often than independence predicts; a negative one means less often, which is the mutual exclusivity that Chapter 3 treats as the signal of interest.

One modification separates $C_{f,f'}$ from the textbook Phi coefficient of a 2×2 table, and it lies in the expectation rather than in the statistic. Taking E_{ij} proportionally to the marginals, as above, assumes that every cell has the same chance of expressing a given feature. Cells differ widely in library size, so the cells that capture less RNA accumulate zeros across all features at once, and a proportional expectation would read that shared under-detection as co-expression and inflate the coefficient. COTAN therefore derives the expected co-occurrence from a model of the zeros themselves: feature g in cell c

is assigned a negative-binomial probability of a zero count with mean $\lambda_g \nu_c$ and shape α_g , where λ_g is the feature's average expression, ν_c scales with the library size of cell c , and α_g is calibrated from the number of zeros actually observed for that feature [1]. Co-expression is thus measured against how often two features' zeros were expected to coincide, not against their raw marginals.

The chi-squared approximation behind $p_{f,f'}$ is asymptotic in the number of cells, not in the counts, and the distinction matters here. With N in the thousands the calibration is reliable for features detected in a reasonable fraction of cells, which covers most of a standard gene matrix. It degrades in the opposite regime, where a feature is rare enough that several expected cell counts E_{ij} fall below about five; Pearson's statistic is then anti-conservative, and such a pair is best read as inconclusive rather than significant. Filtering cannot remove this effect for the rarest transcripts, which is why candidates are ultimately ranked by effect size rather than by p -value alone.

A gene carrying k detected isoforms also contributes $\binom{k}{2}$ pairwise tests, and these are screened at a nominal $\alpha = 0.05$ with no correction applied. Since the tests are counted within genes rather than across the whole matrix, the adjustment a stricter procedure would impose is modest, and the effect-size threshold described in Chapter 3 provides the practical control. A permutation-calibrated null would be more principled and is noted as such in the discussion (Section 4.4).

Output Metrics

The framework reduces the machinery above to three output metrics [1], each of which is used later in this thesis. The first is the co-expression matrix C , whose entries are the coefficients $C_{f,f'}$ just defined; its negative values denote mutual exclusivity, and its positive values co-expression.

The second is differential expression analysis (DEA), which asks whether a single feature f is enriched in a given cell cluster S . It is the same construction applied to a different table: cells inside or outside S are crossed with feature f detected or not, and the resulting correlation coefficient $D_{f,S} \in (-1, 1)$ reports the association. A positive $D_{f,S}$ means f is detected more often inside S than outside it, a negative one less often. Because every DEA coefficient shares one bounded scale, values are comparable across features and across clusters, which is what allows the DTU contrast of Chapter 3 to be formed as the difference of two of them.

The third is the *Global Differentiation Index (GDI)*, a per-feature cluster-specificity measure. For feature i it takes the mean of the smallest 5% of its pairwise p -values against all other features, $\bar{p}_i$, and applies $\text{gdi}_i = \ln(-\ln \bar{p}_i)$, which compresses the distribution of those means into an interpretable range [1]. Constitutively expressed features produce an approximately uniform lower tail and cluster around the null median $\ln(-\ln 0.025) \approx 1.305$, whereas features that vary across cell sub-populations accumulate very small p -values and rise above it. Values beyond roughly 1.5 therefore indicate features worth

following up with DEA, and the GDI is used in Chapter 4 as the check that a clustering has recovered real population structure rather than noise.

Within transcript-level analysis the pairwise coefficient carries a sharper meaning than it does between genes. Observing a strong negative $C_{f,f'}$ between two isoforms of the same parent gene is the primary statistical indicator of alternative transcript usage, since it states that the two are detected in different cells more often than chance allows.

DTU Fundamentals

DTU vs Differential Gene Expression

Differential Transcript Usage (DTU) occurs distinctly when the relative expression proportions of multiple transcripts originating from the exact same parent gene change significantly between different biological conditions or cellular states. This concept differs fundamentally from Differential Gene Expression (DGE), which strictly measures overall changes in total, aggregated gene abundance without regard for internal composition. Biologically, DTU physically reflects shifts in *alternative splicing* decisions—a critical regulatory mechanism enabling cells to produce functionally distinct protein isoforms from an identical genomic sequence. Detecting these precise regulatory shifts inherently requires transcript-level resolution, as aggregating raw counts to the broader gene level mathematically obliterates the nuanced isoform switching signal entirely.

Why Bulk Methods Fail on scRNA-seq

Established bulk RNA-seq DTU methodologies, such as rMATS, SUPPA2, and DRIM-Seq, are structurally engineered to model continuous isoform proportions by relying heavily on deep, internal, junction-spanning read coverage. Consequently, these tools fail catastrophically when applied directly to single-cell data due to three compounding factors [1, 29]. First, the exceptionally low per-cell sequencing depth produces highly degenerate proportion estimates overwhelmingly dominated by zeros. Second, the 3'-bias inherent to droplet protocols renders internal alternative splicing events entirely invisible to traditional junction-spanning algorithms. Finally, pseudo-bulk aggregation recovers the required depth only by discarding the cell-to-cell variation the assay exists to measure [3, 21].

While proportion-based DTU methods have been extended to single-cell paradigms, they remain proportion models: the depth they require is still recovered by aggregation, or depends on junction-spanning coverage the droplet chemistry never provides. As surveyed by Draper et al. [31], none of these contemporary approaches exploits the zero-heavy matrix pattern as a direct source of analytical signal. That gap provides the core motivation for adapting COTAN's sparsity-robust contingency framework to transcript-level analysis.

Mutual Exclusivity Signal

Within the adapted COTAN framework, negative statistical associations—formally defined as *mutual exclusivity*—between features serve as the foundational indicator of alternative usage patterns. When applied specifically to multiple isoforms of the exact same parent gene, this structural mutual exclusivity is biologically consistent with alternative splicing or *isoform switching* events, wherein different cellular states preferentially express entirely distinct transcript variants.

However, mutual exclusivity in isolation is mathematically insufficient to prove a functional switch. Two isoforms may yield a mutually exclusive signature simply because one variant is narrowly confined to a single specific cluster while the other is broadly ubiquitously expressed. This pattern of one-sided enrichment is biologically consistent with tissue-restricted isoform expression, but does not strictly reflect an active regulatory switch in splicing decisions. Identifying a genuine isoform switch requires explicitly demonstrating bidirectional opposition: establishing at least one specific cell cluster in which the first isoform is heavily preferred, and at least one other distinct cluster in which that preference explicitly flips to its partner. This bidirectional contrast serves as the definitive conceptual filter mathematically separating a true *DTU* event from simple one-sided enrichment. While the present chapter successfully establishes mutual exclusivity as the core statistical signal worth testing, the precise mathematical metric that formalises this bidirectional contrast requirement is introduced and explicitly defined in Chapter 3.

Chapter 3

Materials and Methods

This chapter documents every methodological choice made in this thesis. The goal is transparency: an expert should be able to read these decisions, understand the rationale, and judge whether they would make different choices. Reproducibility requires documenting not just what was done, but why alternatives were rejected.

Dataset Selection Strategy

The foundation of any computational analysis is appropriate data selection. Dataset quality bounds every downstream result, since sophisticated algorithms build on the input they receive. The selection strategy prioritized three criteria: known ground-truth biology, *3'-biased* sequencing chemistry (matching real-world constraints), and sufficient cell numbers for statistical power.

Why COTAN Datasets analysis?

Rather than searching the Gene Expression Omnibus (GEO)¹[32] ad hoc, the search started from the curated collection at <https://seriph78.github.io/COTAN.Datasets.analysis/>. This resource offers three advantages. Their quality is pre-validated, since the same collections were used in COTAN's original benchmarking papers, which confirms that they work with the framework and carry no major technical artifacts. The cell line² mixtures have documented identities, so clustering can be checked against expected biology. Many datasets also ship with published *cell barcodes* for high-quality cells, which reduces the risk of including low-quality droplets that would confound downstream analysis.

¹ **Gene Expression Omnibus (GEO):** NCBI's public database of published gene-expression datasets. It stores raw and processed data for re-analysis.

² **Cell line:** A permanently growing population of cells derived from one donor, used as a reproducible experimental system with known biology.

This choice trades breadth (fewer total options than raw GEO search) for reliability—critical when testing whether a novel algorithm actually detects biological signal rather than technical noise. Because the harder case is the informative one, the selection is restricted to 3'-biased droplet data: full-length protocols such as SMART-seq2 make *isoform* distinction straightforward with standard tools, whereas terminal-only coverage leaves many *alternative splicing* events invisible, so success under the droplet constraint establishes a lower bound on COTAN's DTU detection capability and carries practical value for the large-scale studies where cost demands droplet sequencing.

Absence of Published Transcript Matrices

A critical obstacle emerged during the dataset search: **no public scRNA-seq transcript-level count matrices exist**, at least within the COTAN collection. This reflects standard practice in the field rather than an oversight. Most pipelines, including CellRanger³[16] and the Seurat recommended workflow[33], collapse to *gene*-level counts by default; transcript quantification requires additional pseudo-alignment steps with Salmon[26] or alevin-fry[5] that many analysts skip; and even where transcript-level analysis is performed, the results are typically aggregated back to genes before publication.

The consequence is that transcript matrices must be built from the *FASTQ* files directly. This adds computational overhead but also provides complete control over the quantification process—a reproducibility advantage once documented properly.

From the Collection to the Final Datasets

Building those matrices requires a quantification pipeline, and the pipeline imposes a chemistry constraint of its own. `nf-core/scrnaseq`[34] (Section 3.2) handles barcode-based protocols only, with the 10x Genomics[16] 3' chemistries as the supported family. Of the COTAN collection, only the Arigoni human cell-line mixture (GSE243665[35]) and the Ding mouse brain dataset (GSE132044[22]) satisfy both that chemistry requirement and the availability of published barcode lists. Those two datasets are therefore the ones analysed here.

³ **CellRanger / Seurat:** Standard pipelines (10x CellRanger, Seurat) that quantify and analyse scRNA-seq data, defaulting to gene-level counts.

Transcript Matrix Construction Pipeline

With raw sequencing data in hand, the next challenge is choosing tools and documenting modifications required for transcript-level output.

Tool Selection: alevin-fry (simpleaf) vs. kallisto

Two pseudo-aligners dominate single-cell transcript quantification:

Table 3.1: Comparison of scRNA-seq pseudo-alignment tools.

	kallisto – bustools	alevin-fry (simpleaf)
Multi-mapping handling	Uniform distribution	EM-based assignment
UMI deduplication accuracy	Good	Better (modeling errors)
Speed	Fast (~10 min per sample)	Very fast (~5 min per sample)
Integration with nf-core	Available but less common	Native support in v4.2+

alevin-fry (simpleaf mode, part of Salmon) was selected for three reasons. Its UMI handling is superior, since single-cell data relies on *UMIs* to correct *PCR* amplification bias and the alevin-fry error model distinguishes true UMIs from sequencing errors more reliably. Reads compatible with multiple transcripts are assigned probabilistically via Expectation-Maximization (EM)[28] rather than split uniformly, which preserves the relative abundance signals that downstream analysis depends on. Finally, ⁴nf-core/scrnaseq (v4.2.0) supports simpleaf natively, with fewer configuration quirks than the kallisto alternatives.

The EM mode matters: *parsimony-em*⁵ assigns multi-mapping reads to the minimal set of transcripts consistent with the observed data. This produces fractional counts, which are rounded to integers before COTAN ingests the matrix.

The Modification Process (Reproducible Steps)

Standard alevin-fry[5] output collapses to gene level by default via transcript-to-gene mapping⁶. To extract transcript counts, the index files were modified before alignment:

1. Replace t2g.txt with an identity mapping:

```
# Original: ENSMUST00000193814  ENSMUSG000000067548  # gene
# Modified: ENSMUST00000193814  ENSMUST00000193814  # itself
```

⁴ **nf-core:** A community repository of standardised, peer-reviewed Nextflow analysis pipelines.

⁵ **parsimony-em:** An EM mode that assigns each multi-mapping read to the smallest sufficient set of transcripts.

⁶ **Transcript-to-gene (t2g) mapping:** A file pairing each transcript with its parent gene; replacing it with an identity mapping prevents gene-level collapse.

This prevents the final aggregation step that sums isoforms into genes.

2. **Remove** `gene_id_to_names.txt`: The nf-core pipeline expects this file for annotation. Deleting it causes a controlled exit at the annotation stage, after transcript counts have already been written to disk and can be salvaged.

This modification is **reversible**: restoring standard index files returns gene-level quantification without re-running alignment. It adds minimal computational overhead (index editing takes seconds) while enabling transcript-resolution output from a pipeline designed for genes.

Custom Nextflow Pipeline

To make this workflow reproducible across datasets, it was automated with **Nextflow**[36], a domain-specific language for computational pipelines that handles parallelization and checkpointing. The resulting wrapper calls nf-core/scrnaseq (v4.2.0), a community-maintained scRNA-seq analysis pipeline with built-in support for Salmon/simpleaf. This choice provides error handling, containerized dependencies, and standard configuration across datasets.

The custom automation script:

- **Input**: List of SRR accessions⁷ and genome reference (FASTA and GTF⁸).
- **Automated steps**:
 1. Download FASTQ files via `sra-tools`[37].
 2. Build or reuse transcriptome index with `salmon`.
 3. Apply `t2g.txt` modification if transcript-level output is requested.
 4. Execute nf-core/scrnaseq (v4.2.0) with `simpleaf`.
 5. Format outputs for COTAN ingestion (matrix and metadata).
- **GitHub repository**: https://github.com/TODO/GITHUB_LINK. The pipeline includes documentation, example configuration files, and test cases.

This automation reduces the analysis from a multi-day manual process to a single command-line invocation. Future researchers can apply it to new datasets without re-implementing the workflow.

Adapter trimming is omitted deliberately rather than by oversight. The 10x read structure places the barcode and UMI in a fixed header segment, and `simpleaf` resolves that geometry internally with a UMI-aware error model, discarding residual adapter k -mers because they match no indexed sequence. A separate `fastp` pass would therefore reread

⁷ **Sequence Read Archive (SRA)**: NCBI's repository of raw sequencing data, accessed by identifiers such as SRR accessions.

⁸ **GTF (Gene Transfer Format)**: An annotation file listing each gene's coordinates, exon structure, and transcript boundaries; used to build the transcriptome index.

the entire FASTQ file for no change in the count matrix. Quality control at the barcode level still runs: `simpleaf` applies empty-droplet detection to the quantification output, and cells are further restricted to the high-quality barcode sets published with each dataset (Section 3.3).

Data Filtering Choices

Raw count matrices contain noise that must be removed before statistical analysis. The filtering strategy prioritizes biological signal over maximizing cell counts, accepting smaller but higher-quality final datasets.

Cell-level Filters

Cells were filtered using metadata from the original GEO publications. The Arigoni dataset (GSE243665[35]) and Ding dataset (GSE132044[22]) both provide lists of cells that passed their quality control thresholds, and restricting the analysis to those known-good barcodes avoids a second QC pipeline that could introduce inconsistencies. Published studies typically filter on a minimum UMI count per cell, which removes empty droplets; a maximum mitochondrial read percentage⁹, which removes dying cells; and a gene detection threshold, since cells with too few detected genes may be doublets¹⁰ or otherwise low quality. Reusing their filtering ensures comparability and reduces methodological drift.

Final cell counts: 29k human (Arigoni), 4k mouse (Ding).

Feature-level Filters

Transcript matrices contain many features that add noise without signal. *unspliced transcripts* are removed because COTAN's framework[1] was designed for mature mRNA, whereas unspliced pre-mRNA has different statistical properties and can confound *co-expression* analysis; filtering retains spliced-only features and reduces matrix dimensionality. Transcripts detected in fewer than 3 cells across the entire dataset contribute mostly zeros to *contingency tables*, inflating computational cost without improving statistical power, so an ultra-rare isoform cutoff removes them to reduce false positives from spurious associations. After filtering, the human matrix holds 29k cells $\times$ 23k transcripts and the mouse matrix 4k cells $\times$ 12k transcripts.

⁹ **Mitochondrial read percentage:** The fraction of a cell's reads coming from mitochondrial genes; abnormally high values flag dying or damaged cells.

¹⁰ **Doublet:** Two cells accidentally captured inside one droplet, producing a mixed signal that looks like a single abnormal cell.

These choices prioritize quality over quantity. Smaller matrices mean faster computation and cleaner signal—critical when running expensive contingency table calculations across all feature pairs.

Clustering and Validation Strategy

A central question in this thesis is whether transcript-level data captures cellular structure that gene-level analysis misses. To answer this, a three-part clustering strategy was implemented: primary analysis on transcripts, validation against gene-level baselines, and comparison of DTU detection across both approaches.

Transcript-level Clustering (Primary Analysis)

The main analysis clusters cells with COTAN’s hierarchical uniform clustering¹¹. Cell-cell dissimilarity is computed from co-expression patterns, clusters are split recursively until the groups are homogeneous, and adjacent clusters that are statistically indistinguishable are merged. On the mouse dataset the procedure returns 11 uniform clusters plus a small noisy remainder.

This clustering represents the hypothesis: if transcript-level data captures meaningful biological structure, these clusters should correspond to known cell types or states (for datasets with ground truth) or reveal interpretable substructure (for exploratory analysis).

Gene-level Clustering (Validation Baseline)

To assess whether transcript clustering adds value beyond standard approaches, the same cells were clustered independently using gene counts: the official gene-count matrix for the Ding dataset (4k cells $\times$ 28k genes) from GEO, cleaned with the standard COTAN `clean()` function to remove low-quality features and clustered with parameters identical to the transcript analysis, which returns 15 clusters.

This baseline answers: does aggregating transcripts into genes lose information that affects cluster assignment? If both approaches yield similar results, gene-level analysis suffices. If they diverge meaningfully, transcript data captures something unique.

¹¹ **Hierarchical uniform clustering:** COTAN’s procedure that recursively splits cells by co-expression dissimilarity, then merges statistically indistinguishable clusters.

Comparison Approach

The two clustering strategies were compared through three lenses. A confusion matrix¹² cross-tabulates cluster assignments for all 4,077 common cells, where high overlap indicates that both methods capture similar structure and systematic splitting or merging locates the divergence. The DTU detection algorithm was then re-run on transcript-level data with gene-derived cluster assignments as ground truth, which tests whether the signals are robust to how cells are partitioned or depend entirely on clustering resolution. Finally, event overlap analysis compares the genes and transcripts each approach marks DTU-positive: shared events are robust signals surviving both clusterings, transcript-exclusive events are subtle switches requiring fine-grained partitioning, and gene-exclusive events are weaker signals needing aggregated statistical power.

This multi-faceted comparison moves beyond asking which clustering is correct to a more nuanced question: what does each approach reveal, and when should researchers prefer one over the other?

Rationale for the Three-Part Strategy

The validation design addresses three distinct concerns. Biological validity asks whether clusters correspond to meaningful cell types, verified through marker genes and known biology. Methodological contribution asks whether transcript-level analysis adds value beyond standard gene-level approaches, since convergence between the two would leave the extra complexity unjustified. Robustness asks whether the detected isoform switches hold across clustering strategies, or reflect an artifact of a particular partitioning scheme.

Addressing all three concerns establishes whether transcript-level COTAN is both feasible and genuinely useful for DTU detection.

DTU Detection Algorithm

The core contribution of this thesis is an algorithm that identifies *DTU* events from sparse single-cell data using COTAN's statistical framework[1]. Each step is stated here as it is implemented; the biological justification for the two filters is given in Section 2.3.3.

¹² **Confusion matrix:** A table of cluster-label agreement between two clustering runs, revealing systematic splitting and merging.

Step-by-Step Procedure

For each gene with $k \geq 2$ detected isoforms:

1. **mutual exclusivity screening:** For every possible pair of isoforms (A, B) within the gene:
 - Compute the *co-expression* coefficient $C_{A,B}$ from the 2×2 *contingency table* of presence and absence across cells[1] (Section 2.2).
 - Filter for negative $C_{A,B}$ with $p\text{-value}^{13} < 0.05$.
2. **Contrast calculation:** For each significant pair, compute differential enrichment vectors:
$$\Delta_{\text{cluster}} = D_{A,S} - D_{B,S}, \quad (3.1)$$
where $D_{f,S}$ is COTAN's differential expression coefficient for feature f in cluster S (Section 2.2). Each term lies in $(-1, 1)$, so the contrast is itself bounded and comparable across genes and clusters.
3. **Thresholding:** Retain pairs showing isoform switching:
 - At least one cluster with $\Delta > \tau$.
 - At least one cluster with $\Delta < -\tau$.
 - Thresholds:¹⁴ $\tau = 0.2$ (human, well-defined cell lines), $\tau = 0.05$ (mouse, continuous differentiation).

The two filters are ordered deliberately, and the reason they isolate genuine switches rather than one-sided enrichment is argued in Section 2.3.3; this section documents only the procedure itself.

The algorithm outputs a list of gene-isoform pairs with evidence for cluster-specific differential usage, ranked by statistical significance and effect size ($|\Delta|$).

Tuning the Threshold Parameter τ

The threshold parameter τ controls sensitivity versus specificity in DTU detection: high values yield few confident calls but risk missing subtle switches; low values capture more events but increase false positives.

A tuning script was developed at the supervisor's request (Prof. Silvia Galfré). It varies τ across a range such as 0.01 to 0.5, counts the detected DTU events at each threshold, plots the relationship between τ and the number of significant pairs, and suggests thresholds that yield biologically reasonable candidate counts, typically 20–100 events for validation feasibility.

¹³ **p-value:** The probability of observing an association this strong if the features were truly independent.

¹⁴ **τ (contrast threshold):** The minimum absolute DEA contrast between two isoforms required to count a cluster toward a switch.

The script is included in the GitHub repository alongside the main pipeline. Results from this tuning informed the choice of $\tau = 0.2$ for human data (well-separated cell lines, stronger signals) and $\tau = 0.05$ for mouse brain tissue (continuous differentiation trajectory, subtler transitions).

Detailed threshold optimization results appear in Chapter 4 alongside the DTU discovery findings.

Chapter 4

Results and Discussion

This chapter presents the findings from applying transcript-level COTAN to scRNA-seq data. It begins with quality assessment (GDI clustering validation) and a comparison of gene-level and transcript-level structure, before presenting the main contribution: DTU detection results across the human and mouse datasets. Each section builds toward answering whether sparse 3'-biased data can support isoform-switch discovery using contingency-table statistics.

Dataset Quality Assessment via GDI

Before DTU calls can be interpreted, the clustering must be shown to capture biological structure rather than technical noise. The Global Differentiation Index (GDI) provides this validation by measuring cluster homogeneity and marker gene specificity.

Transcript-level Clustering Quality

Figure 4.1 shows the GDI plot for the mouse dataset clustered on transcript counts. Each point represents a cell, colored by its assigned cluster (11 uniform clusters total). The clear spatial separation indicates that transcript-level co-expression patterns distinguish distinct cellular populations.

Key observations from this visualization:

- Cluster boundaries are well-defined with minimal overlap, suggesting robust isoform-based signatures.
- High-GDI genes (marker candidates) show selective enrichment in specific clusters rather than uniform expression.
- The noisy remainder cells (not assigned to any cluster) form a tight central cloud; these were filtered out from downstream DTU analysis.

This confirms that transcript-level data contains sufficient signal for meaningful clustering, despite sparsity and 3'-bias constraints.

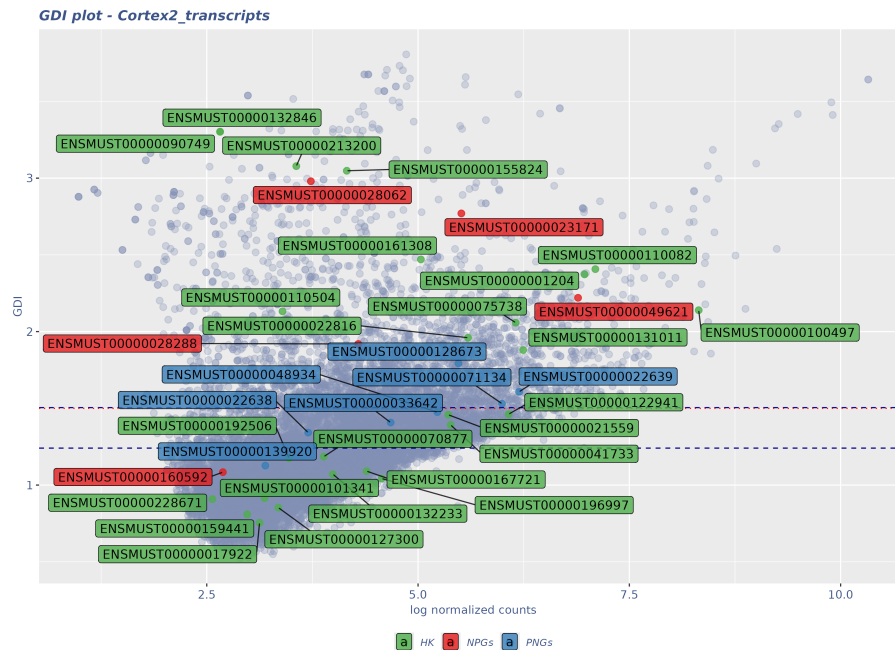


Figure 4.1: GDI plot of the mouse dataset clustered on transcript-level counts, coloured by the 11 uniform clusters.

Gene vs Transcript Structure Comparison

Figure 4.2 shows the same cells clustered using standard gene counts (15 clusters). The different cluster resolution—more numerous but smaller groups compared to transcript clustering—suggests that aggregation to genes loses fine-grained structure.

These GDI analyses establish that both clustering approaches capture real biological signal, but with different sensitivity to cellular substructure. The main question is whether this structural difference translates into distinct DTU detection.

Clustering Comparison: Gene vs Transcript Level

To quantify how gene and transcript clustering diverge, a confusion matrix was constructed across 4,077 common cells present in both analyses.

Figure 4.3 shows the confusion matrix heatmap. Several patterns emerge:

- **Diagonal dominance:** Most cells receive consistent assignments across both methods, validating that neither approach produces random clusters.

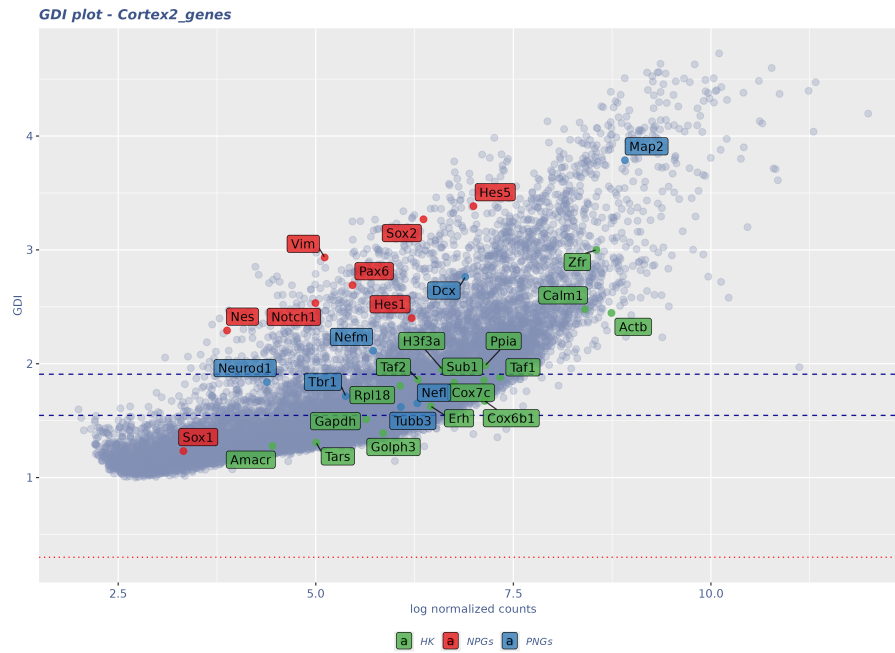


Figure 4.2: GDI plot of the same cells clustered on gene-level counts, yielding 15 clusters.

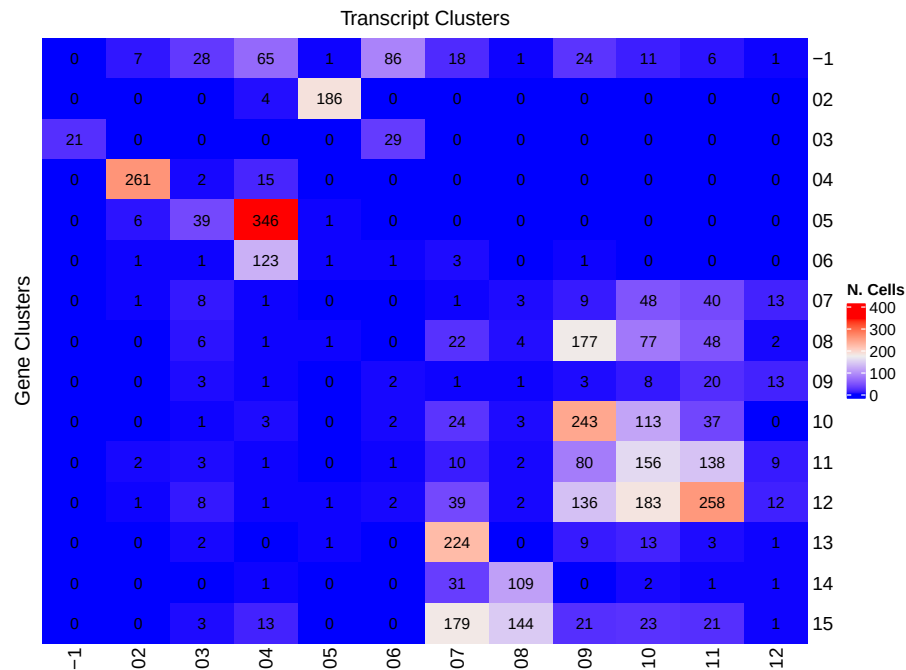


Figure 4.3: Confusion matrix comparing transcript-level (11 clusters) and gene-level (15 clusters) assignments for the 4,077 common cells of the mouse dataset.

- **Systematic splitting:** Several transcript-level clusters correspond to sub-regions of single gene-level clusters. For example, transcript cluster 4 splits from the broader gene cluster 7, suggesting isoform variation within what appears homogeneous at gene resolution.
- **Selective merging:** Conversely, some adjacent transcript clusters merge in gene analysis, indicating that their differences rely on isoform-specific expression patterns rather than overall transcriptional activity.

The interpretation: transcript-level clustering reveals cellular substructure invisible to gene aggregation. This is precisely the regime where DTU detection should add value—if isoform switches distinguish cell states that genes alone cannot separate, then transcript-aware methods capture biologically relevant signals.

However, this raises a critical question: whether the detected DTU events hold across clustering strategies, or depend entirely on how cells are partitioned. The robustness analysis below answers it.

Core Finding: DTU Detection Results

The algorithm from Section 3.5 was applied to both human and mouse datasets, filtering for mutually exclusive isoform pairs with bidirectional cluster preference (threshold τ as specified). The results show that contingency-table statistics can recover within-gene exclusivity patterns even in sparse 3'-biased data.

Human Dataset Findings (Arigoni)

The human cell line mixture provided ground-truth validation through known biology. After thresholding at $\tau = 0.2$:

- **430 genes with significant mutually exclusive isoform pairs** ($p < 0.05$)
- **21 DTU candidates across 9 genes** passing the bidirectional contrast threshold

The higher τ value reflects well-defined cell line clusters with strong separation, allowing stricter filtering for confident calls.

Representative findings:

1. **TPM1 (tropomyosin 1):** Multiple isoforms show opposing preferences across cell lines. Isoform A is enriched in PC9 and DV90; isoform B is dominant in A549 and HCC78.

2. **RPL10, RPS24 (ribosomal proteins)**: Unexpected DTU signals in ribosomal genes suggest cell-type-specific translation regulation—consistent with the heterogeneity of ribosomal composition underlying specialised translation [38].

The relatively small final candidate list (9 genes) reflects the algorithm’s conservative design: requiring both *mutual exclusivity* and bidirectional contrast filters out spurious associations.

Mouse Dataset Findings (Ding)

For mouse brain tissue, a lower threshold ($\tau = 0.05$) was applied to capture subtler transitions along differentiation trajectories:

- **14 genes with significant pairs**
- **46 DTU candidates across 8 genes** after contrast filtering

The larger candidate count despite fewer initial significant pairs suggests that mouse brain tissue exhibits more gradual *isoform switching* patterns rather than sharp cell-line-level switches.

Representative findings:

1. **Meg3 (maternally expressed gene 3)**: Long non-coding RNA¹ with a documented role in neuronal differentiation. Isoform A dominates early progenitor clusters; isoform B is enriched in differentiated neuron clusters, a switch that tracks the differentiation trajectory rather than a sharp binary state.
2. **Malat1 (metastasis-associated lung adenocarcinoma transcript 1)**: Another highly conserved lncRNA involved in splicing regulation. Shows cluster-specific isoform preferences, consistent with the alternative 3′ isoform usage documented for this locus [39].
3. **Srrm2 (serine/arginine-rich splicing factor 2)**: Encodes a core spliceosome² component. Its own DTU suggests feedback regulation—alternative splicing of splicing factors is a known mechanism for tuning isoform landscapes across cell states.
4. **Ttc14 (tetratricopeptide repeat domain 14)**: Novel candidate with no prior DTU literature. Requires *orthogonal validation* to confirm biological significance.

¹ **Long non-coding RNA (lncRNA)**: A long RNA molecule that does not encode a protein but regulates gene expression, splicing, and development.

² **Spliceosome**: The molecular machine that removes introns and joins exons during splicing.

Robustness Analysis: Shared vs Exclusive Events

DTU detection was re-run using gene-derived cluster labels on the same transcript matrix, and the resulting event lists were compared across both clustering strategies:

	Transcript Clustering	Gene Clustering
Total DTU pairs detected	46	46

Table 4.1: DTU event totals under the two clustering strategies (mouse dataset).

Event composition breakdown:

- **Shared events:** Majority of pairs detected by both methods, including *Meg3* and *Malat1*—these represent strong signals surviving regardless of partitioning scheme.
- **Transcript-exclusive (3 pairs):** *Srrm2* and *Ttc14* isoform switches appeared only with transcript-derived clusters. These likely reflect subtle transitions requiring fine-grained population resolution to detect.
- **Gene-exclusive (4 pairs):** *Snrnp70*, *Lrrc7*, *Zfp941*, and a specific *Meg3* pair detected only using gene-level clustering. These may represent weaker signals that benefit from aggregated statistical power.

The interpretation:

1. Strongest DTU signals (high effect size) survive both approaches—validating the algorithm’s core functionality.
2. Subtle switches depend on clustering resolution: transcript-level captures isoform-specific states; gene-level provides power for low-abundance transcripts.
3. Neither approach dominates universally. Researchers should consider their biological question when choosing—fine-grained substructure vs statistical robustness.

This robustness analysis confirms that while the strongest signals are cluster-agnostic, subtle isoform switches reveal themselves only under appropriate partitioning schemes. The choice of clustering strategy is not merely technical—it shapes which biology becomes visible.

Discussion

These findings are now set within their broader methodological and biological context.

Key Findings Interpretation

Two conclusions follow from these results. The top candidates (Meg3, Malat1, TPM1) align with documented biology—lncRNA switching in neurogenesis and ribosomal protein regulation—which indicates that the algorithm recovers genuine signal rather than technical artifacts. Convergence across a human cell line mixture and mouse brain tissue further indicates that the method holds under different biological contexts rather than a single favourable case study.

Limitations and Caveats

Despite encouraging results, several constraints temper interpretation:

- **3'-bias invisibility:** Alternative splicing events occurring in transcript mid-regions remain undetectable when only 3' ends are sequenced. The DTU candidates found here represent a subset—those where isoforms differ at their terminal exons or 3' ends.
- **EM rounding effects:** Fractional EM assignments were rounded to integers for COTAN compatibility. Low-abundance isoforms with small fractional counts may be distorted by this discretization, potentially introducing false negatives or altered contrast values.
- **Threshold sensitivity:** The choice of τ lacks formal statistical grounding. While empirical tuning produced reasonable candidate numbers, a permutation-based null distribution would provide calibrated p -values rather than arbitrary thresholds.
- **Lack of orthogonal validation:** Without long-read sequencing or targeted RT-PCR³ confirmation, false positive rates remain estimated from internal consistency (shared vs exclusive events) rather than ground truth. This is a limitation common to computational DTU studies but remains critical for biological claims.

³ **RT-PCR:** Reverse-transcription polymerase chain reaction: a lab technique that experimentally confirms whether a specific isoform is present, used as an orthogonal check on computational DTU calls.

Future Directions: Extending COTAN for Transcript Analysis

The framework established here opens several avenues for methodological extension:

1. **Integration with long-read data:** Combining short-read depth (thousands of cells) with PacBio/Nanopore⁴ isoform annotation would anchor DTU calls in complete transcript sequences.
This hybrid approach could validate computational predictions while scaling to population-level sample sizes.
2. **Trajectory-aware analysis:** Current clustering assumes discrete cell populations, but differentiation is continuous. Modeling isoform proportions along pseudotime⁵ trajectories (e.g., using COTAN’s framework adapted for regression rather than categorical comparison) could reveal dynamic switching patterns during transitions.
3. **Allele-specific expression extension:** COTAN’s contingency table approach naturally extends to phased variants. Detecting allele-biased isoform usage would connect regulatory genetics (eQTLs, sQTLs)⁶ to splicing outcomes in single cells—bridging population genomics with transcript regulation.
4. **Benchmarking against bulk DTU tools:** Systematic comparison with rMATS, SUPPA2, and DRIMSeq [29, 31] on matched pseudo-bulks would contextualize sensitivity and specificity. This benchmark could identify regimes where single-cell methods add unique value versus when aggregation suffices.

These extensions remain beyond the scope of this thesis but demonstrate that transcript-level COTAN is not a final solution—rather, it establishes proof-of-concept for sparsity-robust isoform analysis in single cells. The framework’s flexibility suggests further adaptations as new data types and biological questions emerge.

⁴ **Long-read sequencing (PacBio, Oxford Nanopore):** Sequencing technologies that produce full-length transcript reads, giving complete isoform annotation unavailable from short 3’-biased reads.

⁵ **Pseudotime:** A one-dimensional ordering of cells along a continuous developmental or differentiation trajectory, capturing gradual transitions rather than discrete populations.

⁶ **eQTL / sQTL:** Genomic variants that measurably affect gene expression (eQTL) or splicing/isoform usage (sQTL); linking them to single-cell isoform switches connects population genetics with regulation.

Chapter 5

Conclusions

This thesis established that COTAN’s sparsity-robust contingency-table framework can be extended to transcript-level analysis, recovering within-gene exclusivity patterns indicative of Differential Transcript Usage (DTU) from sparse 3’-biased single-cell *RNA-seq* data. The key insight—that zero patterns carry reliable signal even in extremely sparse matrices—transfers successfully from gene co-expression to isoform *mutual exclusivity* detection.

Summary of Findings

Applied to the two droplet datasets, the adapted framework returned detectable signal in both: 430 human genes carried a significant mutually exclusive isoform pair, 21 candidates surviving the bidirectional contrast filter, and 46 mouse candidates passed at the lower threshold. The strongest calls—TPM₁ in the cell line mixture, *Meg3* and *Malat1* in the mouse cortex—are consistent with documented isoform biology [39, 38], and most survived the swap of clustering strategy, so the signal does not depend on a particular partitioning scheme.

Broader Implications

This work has implications across three domains:

For computational genomics: The success of contingency-table statistics at transcript resolution demonstrates that sparsity need not be treated as a problem to overcome (via imputation or log-transformation) but rather modeled directly. This philosophy may extend to other single-cell challenges—allele-specific expression, chromatin accessibility peak co-occurrence, and multi-modal integration.

For isoform biology: Detecting DTU in 3'-biased data opens new analysis opportunities for existing large-scale datasets (e.g., Human Cell Atlas, Tabula Sapiens¹) that were previously closed to isoform-level questions. While sensitivity is limited compared to full-length protocols, the trade-off between cell numbers and coverage may favor droplet methods when statistical power from thousands of cells outweighs per-cell depth.

For COTAN framework development: This thesis validates that COTAN's core design principles—direct modelling of zero counts and large- N χ^2 calibration of p -values—generalize beyond gene-level analysis. Future extensions to other single-cell modalities (ATAC-seq² peak co-occurrence, protein abundance correlations in CITE-seq³) appear feasible.

Limitations and Future Work

The constraints on these results—3'-bias invisibility, the absence of orthogonal validation, the heuristic threshold τ , and EM rounding—are set out together with the directions they open in Section 4.4. The same underlying trade-off governs all four: sensitivity is bought with cells rather than coverage, so the value of the method grows as datasets do.

A newer quantifier points the same way. SCALPEL[40] builds the transcript matrix from standard 3'-based data by a different route from the EM step used here, and its authors report higher sensitivity and closer recovery of true isoform proportions than existing tools, with experimental validation. Substituting it for parsimony-em would test whether the downstream DTU calls sharpen, making it a natural extension of this framework.

Final Remarks

This thesis demonstrates that transcript-level COTAN can reveal regulatory programs hidden to gene-centric analyses. The contingency-table framework provides a principled approach to sparse data analysis—one that respects the actual measurement process rather than imposing transformations designed for dense, normally distributed data.

The path forward requires orthogonal validation and the complementary technologies set out in Section 4.4 before isoform-aware single-cell methods move from proof-of-concept to standard practice.

¹ **Human Cell Atlas / Tabula Sapiens:** Large international reference atlases of single-cell transcriptomes across human tissues, relied on 3'-biased droplet data and were therefore closed to isoform-level questions.

² **ATAC-seq:** Assay for Transposase-Accessible Chromatin: maps genome regions where chromatin is open, indicating regulatory activity.

³ **CITE-seq:** A protocol that simultaneously measures RNA and surface proteins in the same single cells, enabling multi-modal analysis.

The framework established here provides a foundation for that future work—combining methodological rigor (contingency-table statistics), computational reproducibility (documented Nextflow pipeline), and biological validation (alignment with known isoform switches). Whether applied to neurogenesis, cancer evolution, or developmental biology, the core question remains: *which transcripts are cells expressing, and how does that choice change across conditions?*

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Appendix A

Supplementary Data

Complete DTU Results (Arigoni and Ding Datasets)

Table A.1: Significant DTU events in the human dataset (Arigoni 2024), including cluster enrichment and maximum contrasts.

Gene	Tr. A	Tr. B	Clusters A	Clusters B	ΔA	ΔB	COEX	p -value
TPM1	ENST00000334895	ENST00000358278	DV90, HCC78, HTB178	A549, CCL-185-IG	0.33	0.61	-0.101	7.72×10^{-67}
RPL10	ENST00000406022	ENST00000458500	A549	HCC78, PC9	0.22	0.46	-0.086	2.78×10^{-49}
RPL10	ENST00000458500	ENST00000491035	HCC78, PC9	A549, PBMCs	0.46	0.25	-0.084	6.41×10^{-47}
TPM1	ENST00000404484	ENST00000651344	HCC78, HTB178	A549, CCL-185-IG	0.27	0.37	-0.074	1.15×10^{-36}
TPM1	ENST00000317516	ENST00000651344	HCC78, HTB178	A549, CCL-185-IG	0.28	0.36	-0.068	4.41×10^{-31}
RPS24	ENST00000360830	ENST00000465692	DV90, HCC78	CCL-185-IG, HTB178, PBMCs	0.28	0.30	-0.051	1.76×10^{-18}
TPM1	ENST00000288398	ENST00000334895	A549, CCL-185-IG	HTB178	0.30	0.21	-0.047	1.13×10^{-15}
TPM1	ENST00000560959	ENST00000651344	HCC78, HTB178	A549, CCL-185-IG	0.22	0.32	-0.044	8.17×10^{-14}
RPS24	ENST00000360830	ENST00000482069	DV90, HCC78	CCL-185-IG, HTB178, PBMCs	0.28	0.28	-0.040	8.15×10^{-12}
TPM1	ENST00000334895	ENST00000651344	DV90, HCC78, HTB178	A549, CCL-185-IG	0.28	0.44	-0.040	8.76×10^{-12}
DSTN	ENST00000449141	ENST00000474024	HTB178	CCL-185-IG	0.28	0.22	-0.039	3.70×10^{-11}
SAT1	ENST00000379253	ENST00000462639	A549	PC9	0.26	0.27	-0.038	6.62×10^{-11}
TPM4	ENST00000645471	ENST00000646974	CCL-185-IG	HCC78	0.32	0.39	-0.034	5.27×10^{-09}
SAT1	ENST00000379251	ENST00000462639	A549	PC9	0.22	0.21	-0.031	7.47×10^{-08}
SAT1	ENST00000379270	ENST00000487713	A549	PC9	0.25	0.25	-0.031	9.35×10^{-08}
RPL35A	ENST00000429437	ENST00000642387	PC9	PBMCs	0.21	0.22	-0.031	1.26×10^{-07}
TPM1	ENST00000558544	ENST00000651344	HCC78, HTB178	A549, CCL-185-IG	0.23	0.33	-0.019	1.04×10^{-03}
TPM1	ENST00000358278	ENST00000558868	A549, CCL-185-IG	CRL5868, DV90, HCC78, HTB178	0.52	0.27	-0.018	1.96×10^{-03}
SNHG5	ENST00000654795	ENST00000664396	CCL-185-IG	PC9	0.23	0.28	-0.018	2.37×10^{-03}
RPS8	ENST00000372209	ENST00000485390	PC9	PBMCs	0.21	0.29	-0.017	4.19×10^{-03}
TPM1	ENST00000317516	ENST00000358278	DV90, HCC78, HTB178	A549, CCL-185-IG	0.33	0.54	-0.012	4.03×10^{-02}

Table A.2: Shared DTU events detected by both clustering methods (Ding dataset), including cell types and contrasts.

Gene	Tr. A	Tr. B	Clusters A	Clusters B	Δ A	Δ B	COEX	p -value
Meg3	ENSMUST00000146701	ENSMUST00000246047	07, 08, 09, 10, 11, 12	−1, 02, 03, 04, 05, 06	0.28	0.66	−0.174	1.58×10^{-28}
Meg3	ENSMUST00000129245	ENSMUST00000146701	−1, 02, 03, 04, 05, 06	07, 08, 09, 10, 11, 12	0.63	0.28	−0.160	1.65×10^{-24}
Malat1	ENSMUST00000173499	ENSMUST00000245771	08, 10, 11, 12	02, 04, 05, 06	0.27	0.29	−0.104	2.91×10^{-11}
Malat1	ENSMUST00000173499	ENSMUST00000245150	08, 09, 10, 11, 12	−1, 02, 04, 05, 06	0.17	0.29	−0.073	3.30×10^{-06}
Meg3	ENSMUST00000129245	ENSMUST00000247387	02, 04, 05, 06	11, 12	0.34	0.29	−0.043	5.70×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000246553	02, 04, 05, 06	11, 12	0.34	0.29	−0.042	6.88×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000245180	02, 04, 05, 06	11, 12	0.33	0.30	−0.042	7.27×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000246499	02, 04, 05, 06	11, 12	0.33	0.29	−0.042	7.40×10^{-03}
Meg3	ENSMUST00000246047	ENSMUST00000247387	02, 04, 05, 06	11, 12	0.37	0.29	−0.042	7.96×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000246427	02, 04, 05, 06	11, 12	0.33	0.29	−0.041	8.84×10^{-03}
Meg3	ENSMUST00000246047	ENSMUST00000246553	02, 04, 05, 06	11, 12	0.37	0.29	−0.041	9.49×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000246700	02, 04, 05, 06	11, 12	0.33	0.29	−0.041	9.51×10^{-03}
Meg3	ENSMUST00000245180	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.29	0.37	−0.041	9.66×10^{-03}
Meg3	ENSMUST00000246047	ENSMUST00000246499	02, 04, 05, 06	11, 12	0.36	0.29	−0.040	9.79×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000143836	02, 04, 05, 06	11, 12	0.33	0.29	−0.040	9.83×10^{-03}
Meg3	ENSMUST00000129245	ENSMUST00000245980	02, 04, 05, 06	11, 12	0.33	0.29	−0.040	1.04×10^{-02}
Meg3	ENSMUST00000246047	ENSMUST00000246427	02, 04, 05, 06	11, 12	0.36	0.29	−0.040	1.15×10^{-02}
Meg3	ENSMUST00000246047	ENSMUST00000246700	02, 04, 05, 06	11, 12	0.36	0.29	−0.039	1.24×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000247194	02, 04, 05, 06	11, 12	0.32	0.29	−0.039	1.26×10^{-02}
Meg3	ENSMUST00000143836	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.28	0.36	−0.039	1.27×10^{-02}
Meg3	ENSMUST00000245980	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.28	0.36	−0.039	1.36×10^{-02}
Meg3	ENSMUST00000246047	ENSMUST00000247194	02, 04, 05, 06	11, 12	0.36	0.28	−0.038	1.60×10^{-02}
Pde10a	ENSMUST00000141877	ENSMUST00000233652	07	09, 10, 11	0.51	0.13	−0.037	1.67×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000243087	02, 04, 05, 06	11, 12	0.32	0.24	−0.034	3.21×10^{-02}
Grin2b	ENSMUST00000053880	ENSMUST00000125905	11	02, 04	0.10	0.10	−0.033	3.39×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000246154	02, 04, 05, 06	11, 12	0.32	0.25	−0.033	3.47×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000243937	02, 04, 05, 06	11, 12	0.31	0.25	−0.033	3.79×10^{-02}
Meg3	ENSMUST00000243087	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.24	0.35	−0.032	3.82×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000143847	02, 04, 05, 06	11, 12	0.31	0.24	−0.032	3.98×10^{-02}

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Gene	Tr. A	Tr. B	Clusters A	Clusters B	Δ A	Δ B	COEX	<i>p</i> -value
Meg3	ENSMUST00000129245	ENSMUST00000247320	02, 04, 05, 06	11, 12	0.31	0.24	−0.032	3.98×10^{-02}
Meg3	ENSMUST00000126289	ENSMUST00000129245	11, 12	02, 04, 05, 06	0.25	0.31	−0.032	4.08×10^{-02}
Meg3	ENSMUST00000246047	ENSMUST00000246154	02, 04, 05, 06	11, 12	0.35	0.24	−0.032	4.12×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000247335	02, 04, 05, 06	11, 12	0.31	0.25	−0.032	4.19×10^{-02}
Meg3	ENSMUST00000243937	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.24	0.35	−0.031	4.45×10^{-02}
Meg3	ENSMUST00000143847	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.24	0.35	−0.031	4.66×10^{-02}
Meg3	ENSMUST00000246047	ENSMUST00000247320	02, 04, 05, 06	11, 12	0.35	0.24	−0.031	4.66×10^{-02}
Meg3	ENSMUST00000129245	ENSMUST00000246462	02, 04, 05, 06	11, 12	0.31	0.24	−0.031	4.73×10^{-02}
Meg3	ENSMUST00000126289	ENSMUST00000246047	11, 12	02, 04, 05, 06	0.25	0.35	−0.031	4.81×10^{-02}
Meg3	ENSMUST00000246047	ENSMUST00000247335	02, 04, 05, 06	11, 12	0.35	0.24	−0.031	4.89×10^{-02}

Table A.3: DTU events exclusive to transcript-level clustering (Ding dataset).

Gene	Tr. A	Tr. B	Clusters A	Clusters B	Δ A	Δ B	COEX	p -value
Srrm2	ENSMUST00000088621	ENSMUST00000186961	10	09, 11	0.25	0.10	-0.059	1.53×10^{-04}
Srrm2	ENSMUST00000088621	ENSMUST00000233636	10	09, 11	0.26	0.12	-0.059	1.78×10^{-04}
Ttc14	ENSMUST00000117915	ENSMUST00000200271	-1	11	0.07	0.06	-0.036	2.19×10^{-02}

Table A.4: DTU events exclusive to gene-level clustering (Ding dataset).

Gene	Tr. A	Tr. B	Clusters A	Clusters B	Δ A	Δ B	COEX	p -value
Snrnp70	ENSMUST00000074575	ENSMUST00000211178	03	14	0.06	0.06	-0.042	6.81×10^{-03}
Lrrc7	ENSMUST00000197866	ENSMUST00000198284	05	08	0.06	0.06	-0.040	9.87×10^{-03}
Zfp941	ENSMUST00000080651	ENSMUST00000106052	12	04	0.08	0.07	-0.032	3.87×10^{-02}
Meg3	ENSMUST00000243087	ENSMUST00000247243	12	-1	0.08	0.06	-0.031	4.89×10^{-02}